

The Hasad-Exotic Matter Interaction Hypothesis (HEMIH)

*A Speculative Hidden-Sector Framework at the Boundary of Islamic Theology,
Quantum Field Theory, and Gravitation*

Speculative research paper • Version 1.0 • September 2026

Epistemic notice: HEMIH is a speculative toy framework. It does not establish that hasad is a physical field, that jinn are exotic matter, or that quantum tunnelling/wormholes explain theological phenomena.

Abstract

This paper develops the Hasad-Exotic Matter Interaction Hypothesis (HEMIH) as a deliberately speculative toy framework. Islamic sources affirm the existence of jinn, discuss hasad (envy), and affirm al-ʿayn (the evil eye), but do not assign these concepts a quantum-mechanical, energetic, or exotic-matter mechanism. Modern quantum field theory (QFT), independently, permits restricted negative renormalized energy densities and quantum transitions, while semiclassical gravity studies geometries whose traversability can depend on violations of classical energy conditions. HEMIH does not identify these established physical ideas with theological entities. Instead, it introduces a hypothetical hasad-associated variable H , a hidden-sector field J , and portal couplings to ordinary matter, then asks what would follow if such a sector existed. A minimal Lagrangian is proposed, three classes of potentially testable predictions are derived, and explicit falsification criteria are stated. The central methodological requirement is separation of theological claims (T), established physics (P), and new hypotheses (H). No empirical evidence presently establishes that hasad generates negative energy, that jinn are exotic matter, or that quantum tunnelling provides a mechanism for supernatural communication.

Keywords: hidden sector; negative energy; quantum energy inequalities; exotic matter; quantum tunnelling; wormholes; hasad; jinn; theology and physics.

1. Epistemic Scope and Method

HEMIH is not offered as established physics and is not presented as an interpretation demanded by Islamic scripture. It is a bridge hypothesis: a mathematical language for asking whether an unknown hidden sector could, in principle, produce measurable effects resembling phenomena described in a theological vocabulary.

Every substantive statement is assigned one of three epistemic classes:

T — Theological claim: a claim grounded in the cited Islamic source.

P — Established physics: a claim supported by mainstream theoretical or experimental physics.

H — HEMIH hypothesis: a new assumption introduced only for exploration.

This separation is essential. A theological statement cannot substitute for laboratory evidence, and a mathematical possibility in QFT cannot establish a theological identification.

2. Theological Boundary Conditions

[T] Qurʾān 113:5 instructs believers to seek refuge “from the evil of an envier when he envies.” The Arabic root ḥ-s-d is the root associated with envy/hasad. The verse does not state that envy emits particles, radiation, negative energy, or any other measurable physical quantity.

[T] Sahih Muslim 2188 reports the statement that the evil eye is real and places it within divine decree. The report does not specify a microscopic mechanism.

[T] Islamic theology also treats jinn as a category of created beings. HEMIH does not claim that scripture identifies jinn with a known field, particle, negative mass, or wormhole-supporting material.

[H] The paper therefore introduces a physical proxy $H(x)$ only as a hypothesis. $H(x)$ is not defined as “hasad itself”; it represents a hypothetical physical correlate that would have to be discovered experimentally, if one exists.

3. Established Physics Relevant to the Analogy

[P] In QFT, renormalized local energy density can be negative relative to a chosen reference state in restricted circumstances. Such results do not imply an unlimited reservoir of negative energy. Quantum energy inequalities constrain suitable averages of negative energy density and are central to evaluating exotic spacetime proposals.

[P] Negative energy density, negative inertial mass, negative gravitational mass, and “exotic matter” are not synonyms.

[P] Quantum tunnelling gives nonzero transition probability across classically forbidden configurations. It does not, by itself, imply that a macroscopic being converts into “pure energy,” traverses arbitrary spacetime distance, and rematerializes.

[P] Traversable wormhole geometries in standard Einstein gravity are associated with violations of classical energy conditions. QFT can violate classical pointwise energy conditions, but quantum energy inequalities impose strong restrictions. Therefore, “QFT permits negative energy” does not entail “usable macroscopic portals exist.”

4. Minimal HEMIH Field Model

Let $H(x)$ be a hypothetical scalar proxy for a hasad-associated physical correlate, and let $J(x)$ be a hypothetical real scalar field representing the simplest possible hidden sector. This choice is intentionally minimal and is not a claim about the ontology of jinn.

Using natural units $\hbar = c = 1$ for the field-theory derivation, define

$$\mathcal{L}_{\text{HEMIH}} = \mathcal{L}_{\text{SM}} + \frac{1}{2}(\partial H)^2 - V_H(H) + \frac{1}{2}(\partial J)^2 - \frac{1}{2}m_J^2 J^2 - (\lambda_J/4!)J^4 - g_{HJ} H J^2 - \kappa J^2 \Box_{\text{SM}}.$$

Here $\mathcal{L}_{\text{SM}}$ is the Standard Model Lagrangian; m_J is the hidden-field mass scale; λ_J is a self-coupling; g_{HJ} couples H to the hidden sector; κ is a weak portal coefficient; and $\Box_{\text{SM}}$ is a Lorentz-scalar Standard Model operator.

A simple H potential is

$$V_H(H) = \frac{1}{2}m_H^2 H^2 + (\lambda_H/4)H^4.$$

The effective quadratic term for J becomes

$$m_{\text{eff}}^2(H) = m_J^2 + 2g_{HJ} H.$$

If $g_{HJ} H < 0$, the effective J mass-squared can be reduced. This is a mathematical consequence of the toy Lagrangian, not evidence that hasad actually changes a physical mass.

5. Localized, Delocalized, and Transition States

The intuitive phrase “mass $\rightarrow$ energy $\rightarrow$ mass” is replaced by a field-theoretic transition:

$$|J_A, \text{localized}\rangle \rightarrow |J^*, \text{delocalized/excited}\rangle \rightarrow |J_B, \text{localized}\rangle.$$

For Hamiltonian $\hat{H}_J$, the transition amplitude is

$$\langle J_B | \exp(-i\hat{H}_J \Delta t / \hbar) | J_A \rangle,$$

with probability

$$P_{A \rightarrow B} = |\langle J_B | \exp(-i\hat{H}_J \Delta t / \hbar) | J_A \rangle|^2.$$

For an ordinary one-dimensional barrier of width L with $E < V$, the WKB tunnelling factor is approximately

$$T \approx \exp[-2L\sqrt{2m(V-E)}/\hbar].$$

HEMIH may hypothesize modified effective parameters in the J sector, but ordinary tunnelling cannot be promoted into arbitrary teleportation without adding new dynamics.

6. Semiclassical Gravity Extension

If the J sector contributes stress-energy, a semiclassical extension can be written

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)[T_{\mu\nu}^{\text{SM}} + \langle T_{\mu\nu}^{\text{J}} \rangle_{\text{ren}}].$$

A candidate “exotic” J configuration would have to be defined by a precise stress-energy property rather than by a label. For example, violation of the null energy condition along a null vector k^μ would mean

$$\langle T_{\mu\nu}(J) \rangle k^\mu k^\nu < 0.$$

This condition is not equivalent to negative mass. Moreover, QFT restrictions on averaged negative energy must be respected. Any HEMIH portal geometry that requires arbitrarily large or long-lived negative energy would conflict with known quantum-energy constraints unless the underlying theory itself modifies those assumptions.

7. Three Quantitative Prediction Classes

Prediction A — Spectral sideband/correlation signature.

Suppose H oscillates or fluctuates around H_0 :

$$H(t) = H_0 + \delta H \cos(\Omega t).$$

Then

$$m_{\text{eff}}^2(t) = m_J^2 + 2g_{HJ}H_0 + 2g_{HJ}\delta H \cos(\Omega t).$$

If J couples weakly to an observable Standard Model channel, the toy model predicts modulation at Ω and potentially harmonics/sidebands. A blinded experiment could search for a pre-registered spectral excess at Ω relative to controls. Failure to observe an effect above a model-specified coupling sensitivity constrains $|kg_{HJ}\delta H|$.

Prediction B — Distance-dependent correlation.

If the effective J excitation has mass m_{eff} , its static propagator in the simplest massive-scalar limit produces Yukawa-like suppression,

$$C(r) \propto e^{(-m_{\text{eff}} r/\hbar c)/r}.$$

Thus a claimed HEMIH-mediated physical correlation cannot remain arbitrarily strong with distance under this minimal model. Measuring no excess correlation across a range of r places a lower bound on m_{eff} or an upper bound on the portal coupling.

Prediction C — Energy-accounting anomaly.

For any closed experimental apparatus,

$$\nabla_\mu T_{\text{total}}^{\mu\nu} = 0.$$

If energy ΔE_{SM} leaves an instrumentally accessible Standard Model channel, the model requires

$$\Delta E_{\text{SM}} + \Delta E_J = 0.$$

A reproducible missing-energy signature would therefore have to exceed calibrated thermal, electromagnetic, mechanical, chemical, and instrumental losses. Conversely, repeated null measurements constrain the rate of energy transfer into J.

8. Operational Null Hypothesis

The primary null hypothesis is

$$H_0: P(X \mid \text{condition}) = P(X \mid \text{matched control}),$$

where X is a pre-defined physical observable.

A HEMIH alternative must specify before data collection:

1. the observable X;
2. the expected sign or spectral/spatial form of the deviation;

3. the measurement bandwidth and sensitivity;
4. exclusion criteria;
5. the statistical threshold;
6. the parameter region that a null result rules out.

Without these commitments, post-hoc reinterpretation can make the hypothesis unfalsifiable.

9. Proposed Experimental Architecture

A scientifically responsible first experiment would not attempt to “summon,” provoke, or harm anyone. It would test only whether a pre-defined, ethically obtained psychological/social condition correlates with an instrument signal beyond chance.

A suitable architecture would use randomized labels, blinded analysis, sham/control sessions, environmental sensors, synchronized clocks, and independent replication. Candidate instruments could include magnetometers, electric-field sensors, microphones, accelerometers, temperature/pressure sensors, radiation counters where appropriate, and quantum sensors only if their expected sensitivity is quantitatively justified.

The protocol must avoid treating subjective experiences as proof of a physical field. Subjective reports may be recorded as a separate variable, but the primary endpoint must be independently measurable.

10. Falsification Criteria

A particular HEMIH parameterization is falsified or strongly disfavored when it predicts an effect larger than experimental sensitivity and repeated, adequately powered, blinded measurements yield the null result.

For example, if the model predicts

$$|\Delta X| \geq X_{\min}$$

with probability at least p^* under a specified condition, while N independent trials achieve sensitivity $\sigma_X \ll X_{\min}$ and show

$$\Delta X = 0 \pm \sigma_X,$$

then that parameter region must be rejected.

HEMIH cannot preserve itself by asserting after every null result that the hidden sector intentionally refused detection. Such a move changes the proposal from a physical hypothesis into an unfalsifiable metaphysical claim.

11. Constraints and Failure Modes

The framework faces severe obstacles. There is no established physical H field associated with hasad. There is no experimental identification of jinn with exotic matter. A scalar J field is merely a modeling convenience. Negative local energy density in QFT is highly constrained. No demonstrated mechanism allows macroscopic portal transport of the type contemplated here. The portal operator $\square_{SM}$ is unspecified until a concrete ultraviolet-complete model is chosen. Human-centered experiments are particularly vulnerable to expectation effects, selective reporting, environmental confounds, and multiple-comparison errors.

These are not minor gaps; they are the central scientific burden of the proposal.

12. Discussion

The strongest form of HEMIH is not the statement “jinn are exotic mass.” That statement collapses several distinct concepts and has no present evidential basis. A more coherent formulation is conditional:

If an unknown hidden sector J exists; if some measurable physical correlate H accompanies the theological/social phenomenon being studied; if H couples to J ; and if J couples weakly to ordinary matter, then a field-theoretic model can be written and constrained experimentally.

This conditional structure makes the framework scientifically useful even if the final result is negative. Experiments can exclude couplings, masses, ranges, or modulation strengths without claiming to test theology itself.

13. Conclusion

HEMIH provides a disciplined mathematical language for a speculative idea while preserving a strict boundary between revelation, established physics, and new conjecture.

The compact interaction chain is

$$H \rightarrow J^* \rightarrow J_{\text{delocalized}} \rightarrow J_{\text{localized}} \rightarrow \square_{\text{SM}},$$

but every arrow is hypotheticalal.

The minimal toy model

$$\mathcal{L}_{\text{HEMIH}} = \mathcal{L}_{\text{SM}} + \mathcal{L}_H + \mathcal{L}_J + \mathcal{L}_{\text{portal}}$$

allows concrete consequences to be derived: spectral modulation, finite-range correlations, and energy-accounting constraints. These predictions make selected parameterizations falsifiable.

At present, there is no empirical basis for concluding that hasad generates negative energy, that jinn consist of exotic matter, or that quantum tunnelling or wormholes mediate communication with humans. HEMIH should therefore be treated as speculative theoretical exploration unless and until reproducible evidence distinguishes it from ordinary physical and statistical explanations.

References

- [1] The Qur'ān, Sūrat al-Falaq 113:5. Translation comparison and morphology: Quranic Arabic Corpus; legacy Quran.com.
- [2] Sahih Muslim 2188, Book of Greetings: report concerning al-‘ayn (the evil eye).
- [3] Fewster, C. J. (2012). Lectures on Quantum Energy Inequalities. arXiv:1208.5399.
- [4] Fewster, C. J., & Eveson, S. P. (1998). Bounds on negative energy densities in flat spacetime. *Physical Review D*, 58, 084010. DOI: 10.1103/PhysRevD.58.084010.
- [5] Ford, L. H., Pfenning, M. J., & Roman, T. A. (1998). Quantum inequalities and singular negative energy densities. *Physical Review D*, 57, 4839. DOI: 10.1103/PhysRevD.57.4839.
- [6] Fewster, C. J., & Roman, T. A. (2003). Null energy conditions in quantum field theory. *Physical Review D*, 67, 044003. DOI: 10.1103/PhysRevD.67.044003.
- [7] Kontou, E.-A. (2024). Wormhole Restrictions from Quantum Energy Inequalities. *Universe*, 10, 291. DOI: 10.3390/universe10070291.
- [8] Standard references on quantum field theory and semiclassical gravity should be added if HEMIH is developed beyond this toy-model stage.